



A randomized controlled trial comparing bone mineral density changes of three different ACL reconstruction techniques^{☆,☆☆}

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ABSTRACT

Introduction: This study aimed to compare the changes in bone mineral density (BMD) of three different ACL reconstruction (ACLR) techniques and its association with early clinical and functional outcomes.

Methods: Sixty-two male adult patients undergoing primary ACLR were prospectively parallel randomized to bone–patellar tendon–bone graft (BPTB), single-bundle (HT-SB) or double-bundle (HT-DB) hamstring graft. BMD (primary outcome) at the proximal tibia, distal femur, femoral neck and trochanteric region was measured blindly at day 1, 3 months, 5 months and 1 year after surgery. KT-1000, Lysholm, IKDC, one-leg hop test and Lachman test were performed blindly at baseline and 1 year post-reconstruction.

Results: There was a significant bone loss at the injured knee and hip at 3 and 5 months which was reversible at the knee, but not at the hip, at 1 year post-operation. There was a significant improvement of early clinical and functional outcomes at 1 year. No significant differences in bone loss was detected among different surgical techniques, except BMD loss at the femoral neck, though a trend of greater BMD loss in the HT-SB group at 5 months after reconstruction was observed. There was a significant positive correlation between BMD at the distal femur and the single-leg hop distance at 1 year.

Conclusion: In conclusion, the three surgical techniques were similar in transient bone loss at the knee region, irreversible bone loss at the hip, early clinical and functional outcomes up to 1 year post-reconstruction. BMD at the distal femur was positively associated with the single-leg hop distance at 1 year post-reconstruction.

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1. Introduction

Anterior cruciate ligament (ACL) reconstruction requires passing a tendon graft through a bone tunnel. Autologous bone–patellar tendon–bone (BPTB) and hamstring (HT) grafts are presently the most commonly used grafts for ACL reconstruction. Adequate bone mass is essential for soft tissue graft fixation to bone [1,2]. Significant bone loss and decreased mechanical properties after flexor tendon insertion site injury and repair were reported, supporting the relationship between bone loss and strength [1,2].

Recent clinical studies showed that there was a significant decrease in bone mineral density (BMD) around the knee region after

ACL reconstruction with BPTB graft and the decrease could persist for many years without returning to the baseline level [3–5]. However, these were not randomized or controlled clinical studies [3–5]. The clinical relevance of post-operative bone loss remains unclear and evidences were inconsistent [3,4,6]. The decrease in bone stock might negatively affect early tendon graft to the bone tunnel healing and hence early stability of knee joint post-reconstruction. Early stability of knee joint allows the patients to adopt more aggressive rehabilitation therapy. Therefore, it would be interesting to understand the impact of BMD on the early functional or clinical outcomes.

Of all the clinical studies reporting BMD loss post-reconstruction, BPTB grafts were used [3–5]. It is not known whether bone loss is seen after ACL reconstruction with the use of single bundle hamstring grafts (HT-SB) and double bundle hamstring (HT-DB) grafts. It is known that soft tissue-to-bone healing occurs slower than bone-to-bone healing [2,7] and bone-to-bone healing occurs in BPTB while tendon-to-bone healing happens in HT-SB and HT-DB. ACL reconstruction with BPTB, HT-SB and HT-DB grafts might induce different adaptive bone changes and hence different degrees of BMD loss due to the different extent of trauma induced, different healing responses, as well as different numbers, locations and angles of tunnels which act as stress risers and has a shielding effect on loading [8,9].

[☆] Clinical trial registration: This study was also registered with the <http://www.clinicaltrials.gov> in the United States (Clinical Trial ID: NCT00418964).

^{☆☆} We followed the CONSORT statement 2010 in reporting the results of this study.

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Our hypotheses were set as follows: (1) Different surgical techniques would induce different degrees of bone loss at the knee region and (2) Bone loss at the knee region might negatively impact the early functional and clinical outcomes of patients. Therefore, in this study, we aimed to compare the changes in BMD at the knee and hip, of three different ACL reconstruction techniques, including BPTB, HT-SB and HT-DB and the relationship of BMD with early clinical and functional outcomes at 1 year in a randomized controlled clinical study.

2. Methods

2.1. Trial design

This is parallel design with 1:1:1 allocation ratio. There is no important change to methods after trial commencement.

2.2. Subject recruitment and randomization

This study was approved by the Clinical Research Ethics Committee of the authors' institution (ref no: CRE-2006.417). We followed the Declaration of Helsinki and ICH-GCP E6 guidelines during the conduct of this study. This study was also registered with the <http://www.clinicaltrials.gov> in the United States (Clinical Trial ID: NCT00418964).

Sixty-two male adult patients undergoing primary ACL reconstruction and meeting the inclusion and exclusion criteria were recruited from the affiliated hospital of the author's institution from Dec, 2007 to Jan, 2009 by the same investigator who performed the surgery. The inclusion and exclusion criteria were shown in Table 1. All patients signed the consent form for participating in this study. Surgical operation, rehabilitation and follow-up assessments were all done in the affiliated hospital of the author's institution.

Randomization was performed by one of the investigators who did not participate in the recruitment of patients. Block randomization with block size of six was applied to divide the patients into three groups: bone patellar tendon bone (BPTB), single bundle hamstring (HT-SB) and double bundle hamstring (HT-DB). Block sequences were randomly selected from all possible combinations (90 combinations in total with block size of six) listed out using computer program. Two patients will be assigned to each group in each block according to the order in the randomly-selected block. The use of block randomization has a better guarantee of equal numbers of subjects in each group when subject characteristics may change over time as a result of change in recruitment policy or when the sample size is small. It is also good for interim analyses as the numbers of

subjects are more or less the same at any point during the recruitment process.

2.3. Surgical techniques

The same experienced surgeon performed all the surgical operations for the three surgical techniques according to the randomization list. It is a common practice in our hospital to use double fixation devices. Blinding of patients and the investigators was considered not practical in this study.

For BPTB surgery, the central one-third of patellar tendon (10 mm in width) was harvested with one end over the apex of the patella and the other just above the tibial tubercle. The femoral tunnel (9–10 mm in diameter) was prepared with the use of an offset guide inserted through the medial portal. The tibial tunnel (9–10 mm in diameter) was prepared with a 55° tibial jig. The graft was then inserted with the aid of a guide pin with an eyelet and was fixed with interference screws over the femur and tibia.

For HT-SB surgery, the semitendinosus and gracilis tendons (about 7–9 mm in diameter) were harvested and inserted into the femoral and tibial tunnels (both about 7–9 mm in diameter) similar to that in BPTB surgery. Graft was double fixed to the tunnel with interference screw and Endobutton at the femoral side and fixed with interference screw and staple at the tibial side.

For HT-DB surgery, the semitendinosus and gracilis tendons were harvested. Two tunnels (6–7 mm in diameter for anteromedial (AM) tunnel and 5–7 mm in diameter for posterolateral (PL tunnel)) were drilled over both the femur and tibia. Both the PL and AM tibial tunnels were prepared with a 55° and 45° tibial jig, respectively. Graft was inserted with the aid of a guide pin with an eyelet. Both the PL and AM grafts in the femoral tunnels were fixed with interference screws and Endobuttons whereas they were fixed with interference screws and staples in the tibial tunnels.

2.4. Postoperative rehabilitation protocol

Postoperatively, all patients underwent the same rehabilitation protocol provided by the Physiotherapy Department of the affiliation hospital of the author's institution. The protocol encourages early weight bearing, full range of knee motion as well as quadriceps strengthening. No brace or splints were used. Progression was made on tolerated basis as guided by pain and swelling. Agility drill including running forward, backward, sideway jump and stairs walking was permitted at 4 months. Contact sports were permitted at 6–9 months, provided that the patient regained full functional stability. Compliance to the rehabilitation protocol in terms of attendance was recorded at each follow-up session. Each patient was expected to attend >40% of total sessions offered or at least five sessions within the first 3 months post-operation (considering standard rehabilitation protocol of 3 months, with one session provided per week).

2.5. Assessment

Demographic information and BMD were assessed at baseline (within 2 weeks pre-operation)/post-op day 1 (for BMD only), month 3, month 5 and 1 year by an experienced blinded technician. The clinical (manual Lachman test, KT-1000) and functional (single-leg hop test, Lysholm score, IKDC score) outcomes of patients were assessed at baseline (within 2 weeks pre-operation) and 1 year by a blinded physiotherapist. Details of the tests are described below.

For BMD measurement, the local bone mineral density (g/cm²) [2]) at the bilateral distal femur, proximal tibia, femoral neck and trochanteric region of each patient was measured by an experienced blinded technician using DXA Norland XR-36 DXA scanner (Norland Inc., Fort Atkinson, WI, USA) [10,11]. Lateral knee scan included the distal femur and proximal tibia, with 8 cm above and 8 cm below the joint

Table 1
Inclusion and exclusion criteria for patient recruitment in this study.

Inclusion criteria	Exclusion criteria
1. Men aged 18–40 at the time of surgery;	1. ACL injury less than 6 weeks;
2. First ACL reconstruction surgery;	2. Injury on duty cases;
3. Single leg involvement; and	3. Other associated injuries (fractures and other ligament involvement such as neurovascular bundles injury);
4. Able to attend pre-operative assessment.	4. Chondral lesion with concomitant intervention;
	5. Concomitant meniscus repair in the same operation, or within 3 months before the operation;
	6. Presence of X-ray features of osteoarthritis including decrease in joint space, presence of osteophytes and subchondral cysts; and
	7. Known chronic disease or receiving long-term medications affecting bone metabolism; and 8. Neurological deficit.

line. Patients were supported in side lying (right side lying for measurement of right knee and left side lying for measurement of left knee) with extended knee. For the quantification of BMD at the distal femur, a rectangle of 4 cm in height rising above the horizontal joint reference line and with the width covering the widest part of the femur was selected as the region of interest (Fig. 1A). For the calculation of BMD at the proximal tibia, a rectangular region of 4 cm in height and with the width covering the tibial tuberosity before overlapping with fibula in the lateral DXA image was located below the horizontal reference line at the knee joint (Fig. 1A). The screw was digitally excluded on the image based on the post-operative X-ray images and the higher signal intensity of the screw shown in the DXA image. For the measurement of BMD at the femoral neck and the trochanteric region, the patient was in supine position and the BMD at both sites was calculated automatically by the program (version 2.3.1) of the DXA machine (Fig. 1B). The scanner has a scanning resolution of 1.0 mm × 1.0 mm and was calibrated before the first measurement on each day with a lumbar spine phantom according to the manufacturer's recommendations. The precision, expressed as coefficient of variation (CV), of the phantom measurement was 1.0%. The CV for BMD measurement at the trochanteric region and femoral neck in patients was 2.5% and 2.1%, respectively, in our laboratory. The % BMD change compared with day 1 was calculated with the formula: $(\text{BMD}_{\text{time point}} - \text{BMD}_{\text{day1}}) / \text{BMD}_{\text{day1}} \times 100\%$. As a result, bone loss due to tunnel drilling was canceled and BMD changes could be compared among different groups.

The IKDC knee form 2000 was used to assess both objective and subjective symptoms, function and sports activity of patients based on self-report [12,13]. A score was obtained.

The single-leg hop test measures the functional stability of the knee [14]. The patient was asked to perform single-leg hop on both the injured and non-injured legs three times and the hop distance was recorded. An index score, calculated by dividing the averaged hop distances of the injured leg by that of the non-injured leg, was reported.

The Lysholm score was used to assess the subjective functional and psychological statuses of patients [15]. A score in the range of 1–100 in ordinal scale was reported.

The anterior laxity of the knee was assessed using KT-1000 (MEDmetric Corp., San Diego, CA). The patient was asked to lie supine and the anterior displacement of tibia (mm) at an applied load of 30 lbs was recorded [16]. The side-to-side difference in anterior laxity was calculated by subtracting the KT-1000 value of the non-injured knee from the injured knee. A higher value in side-to-side difference indicates higher anterior laxity.

The manual Lachman test was used to assess the anterior knee laxity and was graded into four different laxity levels. 0: No laxity; 1: <5 mm anterior translation; 2: 5–10 mm anterior translation; 3: > 10 mm anterior translation.

2.6. Sample size calculation

The primary objective of this study was to detect the % BMD loss around the knee among different surgical techniques. The % BMD loss compared to baseline at month 5 was expected to have the biggest BMD loss and hence was used as the time for the comparison of BMD loss among different surgical techniques. One-way ANOVA was used to compare the % BMD loss among different surgical techniques. Our pilot study showed that the standard deviation of % BMD loss of patients was about 5%. Assume $1-\beta=0.8$, $\alpha=0.05$ and the hypothesized mean % BMD loss at month 5 equaled to 6%, 10% and 4% for BPTB, HT-SB and HT-DB, respectively, 14 patients per group were required to observe the difference in the final statistical analyses. Assuming an attribute rate of 40%, at least 20 patients were recruited in each group. Sample size calculation was done using Power Analysis and Sample Size 2008 software (PASS2008, NCSS Statistical Software, UT, USA).

2.7. Data management and statistical analysis

Patients would be excluded if there was a serious violation of protocol including (i) violation of exclusion criteria found in intra-operation; (ii) lost to follow up of two or more assessment sessions; (iii) failure to attend at least 40% (i.e. five out of the expected 12 sessions) of standard physiotherapy service within the first 3 months in our hospital and (iv) others (such as inadequate graft length and patient's choice of operation). Missing data during follow-up was handled by carry-forward of the results from the previous session.

Descriptive data was presented as mean (standard deviation (SD)) for data in interval/ratio scales, median and range for data in ordinal scale as well as counts and percentages for data in nominal scale. Comparability of baseline demographic information among patients in different groups was done using one-way ANOVA followed by post-hoc pair wise comparison with Bonferroni correction for data in interval/ratio scale, Kruskal–Wallis test followed by Mann–Whitney *U* test with Bonferroni correction for data in ordinal scale and Pearson's chi square exact test for data in nominal scale.

The effects of surgical procedures and time on the % change in BMD, side-to-side difference as measured by KT-1000, index score

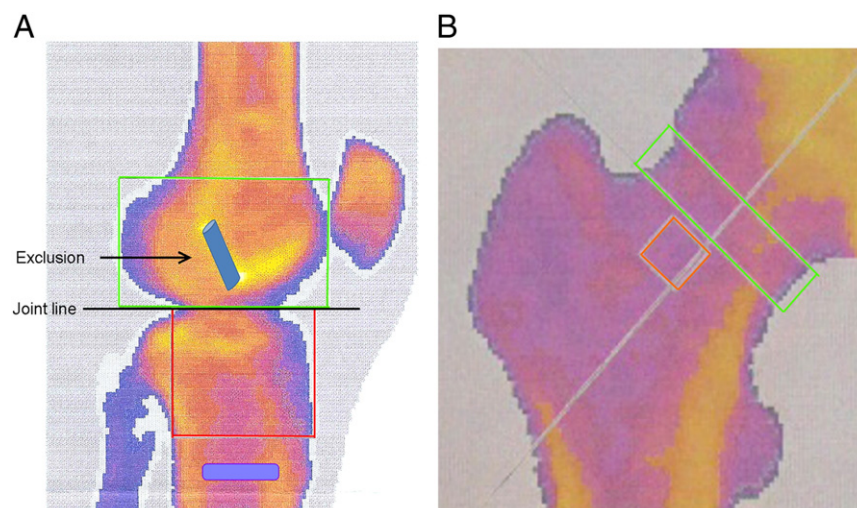


Fig. 1. DXA images showing the region of interest (ROI) for the calculation of BMD at the (A) distal femur (box region) and proximal tibia (box region) as well as (B) trochanteric region (square box region) and femur neck (rectangular box region). The screws (cylinders in panel A) were excluded digitally on the DXA image before BMD measurement.

for one-leg hop test and IKDC score were analyzed by repeated measures of ANOVA, followed by post-hoc pair wise comparison among different surgical procedures or time points with Bonferroni correction for significant overall differences. The manual Lachman test score and Lysholm score at baseline and 1 year were analyzed by the Wilcoxon signed rank test. The differences in Lachman test score and Lysholm score at baseline and at 1 year among the three surgical procedures were compared with Kruskal–Wallis test followed by Mann–Whitney *U* test. The data for BMD, clinical and functional outcomes of different groups were combined for analysis of the changes over time if there was no significant difference among different surgical procedures. Correlation analyses of absolute BMD value at the injured knee with absolute values of functional and clinical outcomes at 1 year were done by Pearson's correlation or Spearman's rho tests. All the data analysis was performed using SPSS analysis software (Version 16.0, ©2005 SPSS Inc.). The level of significance was set at $p \leq 0.050$, 2-tailed.

3. Results

Eighty-two patients were assessed for eligibility from Dec, 2007 to Jan, 2009. Twelve patients did not meet the inclusion criteria and eight patients declined to participate in the study. As a result, 62 patients were recruited for this study. After random allocation, 20, 22 and 20 patients were assigned to the BPTB, HT-SB and HT-DB groups, respectively. One patient (BPTB group) has violated randomization upon surgery due to patient's preference for surgery and was excluded. Another nine patients (BPTB: 5; HT-SB: 1; HT-DB: 3) were also excluded because they have attended less than two follow-up assessments. Seven out of nine (BPTB: 5; HT-DB: 2) of these patients did not attend the follow-up assessments because of work. In fact, three patients in BPTB

group and one patient in HT-DB group have not attended a single session, including the baseline assessment, after enrollment because of work. We were unable to contact the other two patients and hence the reason for loss to follow-up was unknown. None of the remaining patients failed to compile to our rehabilitation program. As a result, 52 patients with 14, 21 and 17 in BPTB group, HT-SB group and HT-DB group, respectively have attended at least three out of four sessions of assessments and also complied with the rehabilitation program. A flowchart showing the flow of patients was shown in Fig. 2. The baseline demographic characteristics, post-op day 1 BMD values and pre-op clinical and functional outcomes of patients in different groups were comparable except for patients' age ($p = 0.050$) (Table 2). It was not surprising given the number of tests that have been performed. Moreover, subsequent post-hoc pair wise comparison of different groups with Bonferroni correction showed no significant difference among different groups in age (BPTB versus HT-SB: $p = 0.122$; HT-SB versus HT-DB: $p = 0.106$; BPTB versus HT-DB: $p = 1.000$).

BMD at both the distal femur and proximal tibia decreased significantly at month 3 post-op (distal femur: -6.5% (6.9%); proximal tibia: -4.8% (7.0%)) and month 5 post-op (distal femur: -7.3% (10.1%); proximal tibia: -5.6% (12.6%)), increased and became insignificantly different from day 1 at 1 year post-op (distal femur: -1.6% (10.3%); proximal tibia: 0% (13.0%)) (both value for all groups: $p < 0.001$) (Fig. 3A). There was no significant difference in BMD loss at the distal femur ($p = 0.205$) and proximal tibia ($p = 0.121$) among the three surgical techniques though a trend of greater loss with the HT-SB group was observed (Fig. 3B and C). BMD loss continued at month 5 for the HT-SB group at the distal femur while the decrease in BMD in the BPTB and HT-DB groups stopped and returned toward the baseline (Fig. 3B). At the proximal tibia, the decrease in BMD continued from month 3 to month 5 in the BPTB and HT-SB groups but returned toward baseline from month 3 to month 5 in the HT-DB group (Fig. 3C).

At the injured hip, BMD at both the trochanteric region (-3.6% (3.3%), -4.5% (8.2%), -4.2% (6.2%) for month 3, month 5 and 1 year, respectively) (value for all groups $p < 0.001$) and femoral neck (-1.9% (3.7%), -2.4% (8.3%), -2.5% (6.8%) for month 3, month 5 and 1 year, respectively) (value for all groups $p = 0.003$) showed significant decrease with time and did not return to baseline at 1 year post-op (Fig. 3D). While there was no significant difference in BMD loss among the three surgical techniques at the trochanteric region (value for all groups $p = 0.153$) (Fig. 3E),

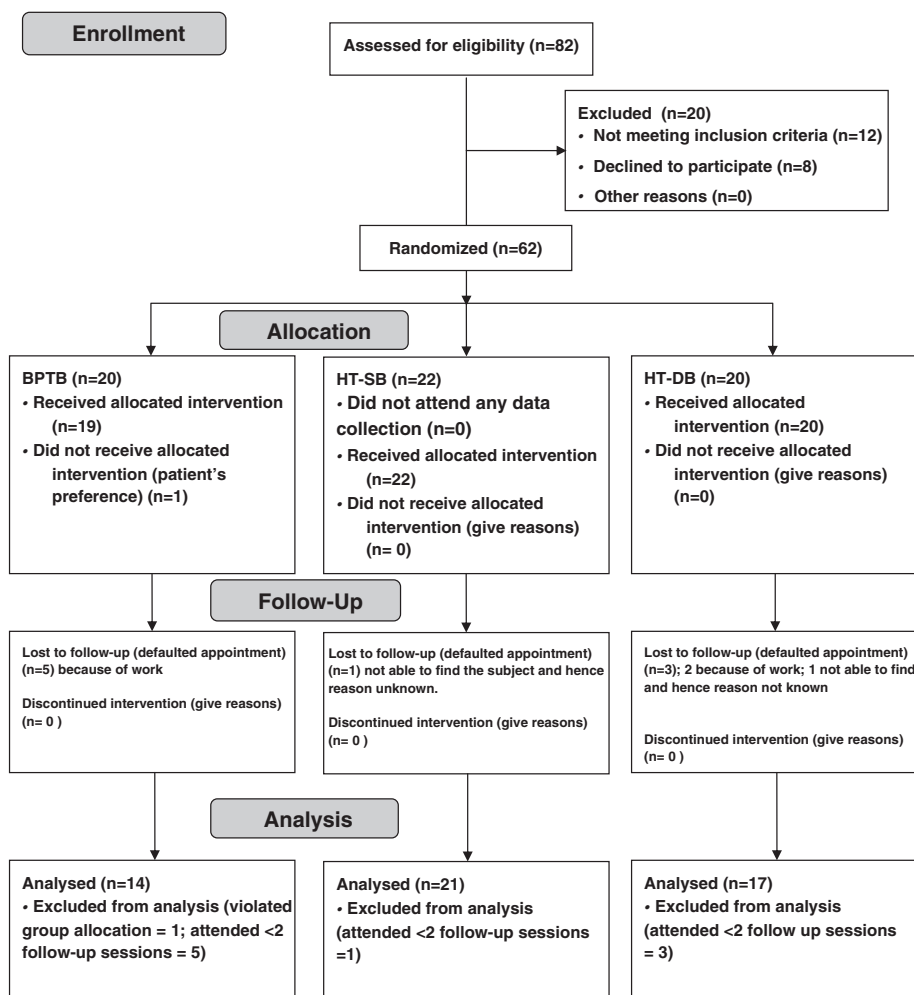


Fig. 2. A flowchart showing the flow of patients in the study.

Table 2

A and B. Demographic characteristics, post-op day 1 BMD values and pre-op clinical and functional outcomes of patients undergoing ACL reconstruction with BPTB, HT-SB or HT-DB.

A		BPTB (n = 14)	HT-SB (n = 21)	HT-DB (n = 17)	p value
Age (years)		26.5 (5.0)	23.2 (3.9)	26.4 (4.9)	0.050
Height (cm)		172.0 (6.3)	172.4 (5.4)	175.9 (5.9)	0.214
Missing		2	4	6	
Weight (kg)		69.8 (10.6)	67.6 (9.9)	70.8 (9.6)	0.651
Missing		1	3	2	
Average time before surgery (months)		7.2 (3.2)	11.6 (11.8)	8.9 (5.2)	0.284
Missing		1	1	2	
Average time to off crutches (days)		21.3 (14.6)	17.6 (13.9)	21.1 (11.4)	0.636
BMD distal femur (g/cm ²)					
Injured		1.32 (0.15)	1.41 (0.16)	1.37 (0.18)	0.282
Non-injured		1.44 (0.16)	1.52 (0.15)	1.49 (0.18)	0.349
BMD proximal tibia (g/cm ²)					
Injured		1.27 (0.21)	1.20 (0.14)	1.19 (0.19)	0.412
Non-injured		1.27 (0.17)	1.31 (0.16)	1.30 (0.19)	0.798
BMD trochanteric region (g/cm ²)					
Injured		0.79 (0.11)	0.82 (0.10)	0.83 (0.12)	0.592
Non-injured		0.84 (0.11)	0.85 (0.12)	0.87 (0.12)	0.770
BMD femoral neck (g/cm ²)					
Injured		1.00 (0.14)	1.00 (0.13)	1.04 (0.12)	0.719
Non-injured		1.04 (0.15)	1.04 (0.14)	1.05 (0.12)	0.990
One-leg hop test					
Injured (cm)		81.8 (50.4)	82.2 (42.2)	72.6 (50.4)	0.800
Non-injured (cm)		127.0 (37.2)	129.6 (28.7)	132.9 (22.1)	0.859
Injured/non-injured		0.6 (0.3)	0.6 (0.3)	0.5 (0.4)	0.684
Missing		1	1	0	
Lysholm score		84 (54–100)	87 (58–100)	80 (43–99)	0.161
Missing		1	0	0	
IKDC		71.9 (15.2)	76.5 (13.3)	73.1 (12.3)	0.583
Missing		1	0	0	
KT-1000 (30 lbs) (mm)					
Injured		10.8 (2.2)	9.1 (2.6)	9.1 (3.1)	0.152
Non-injured		7.1 (2.2)	5.0 (2.2)	6.0 (2.7)	0.051
Side-to-side difference		3.7 (1.4)	4.1 (1.6)	3.1 (1.6)	0.141
Missing		1	0	0	
Lachman test		4 (2–4)	3 (2–4)	3 (3.4)	0.592
Missing		1	0	0	
B		BPTB (n = 14)	HT-SB (n = 21)	HT-DB (n = 17)	p value
Smoking status	Non-smoker	11 (79%)	18 (86%)	15 (88%)	0.725
	Smoker	1 (7%)	1 (5%)	0 (0%)	
	Missing	2 (14%)	2 (10%)	2 (12%)	
	Total	14 (100%)	21 (100%)	17 (100%)	
Premorbid activity level	Elite	1 (7%)	3 (14%)	2 (12%)	0.872
	Recreational	13 (93%)	18 (86%)	15 (88%)	
	Missing	0 (0%)	0 (0%)	0 (0%)	
	Total	14 (100%)	21 (100%)	17 (100%)	
Mechanism of injury	Jump and land	4 (29%)	4 (19%)	4 (24%)	0.862
	Twisting	4 (29%)	8 (38%)	6 (35%)	
	Jump and land with twisting	1 (7%)	5 (24%)	3 (18%)	
	Others	4 (29%)	3 (14%)	3 (18%)	
	Missing	1 (7%)	1 (5%)	1 (6%)	
	Total	14 (100%)	21 (100%)	17 (100%)	

Values for Lysholm score and Lachman test were presented as median (range) while values for the other parameters were represented as mean (SD).

significant difference was detected at the femoral neck (value for all groups $p = 0.043$) (Fig. 3F). However, subsequent post-hoc pair-wise comparison showed that the higher BMD loss in the HT-SB group compared to that in the HT-DB group was not statistically significant (post hoc test $p = 0.054$) (Fig. 3F). Similar to the cases at the distal femur and proximal tibia, greatest BMD loss occurred in the HT-SB group at month 5 post-op (Fig. 3E and F). There was no significant BMD loss at all the measurement sites at the non-injured limb with time (values for all groups: $p = 0.836, 0.952, 0.082, 0.073$

for distal femur, proximal tibia, trochanteric region and femoral neck, respectively) and there was no significant difference among different surgical techniques (values for all groups: $p = 0.190, 0.069, 0.558, 0.269$ for distal femur, proximal tibia, trochanteric region and femoral neck, respectively) (results not shown).

There was no significant difference in IKDC score ($p = 0.759$), Lysholm score ($p = 0.541$), single-leg hop distance ratio ($p = 0.504$), KT-1000 side-to-side difference ($p = 0.120$) and manual Lachman test score ($p = 0.477$) among different surgical techniques at 1 year post-reconstruction. Hence we combined the data from different groups. There was significant improvement in IKDC score, Lysholm score, single-leg hop distance ratio, KT-1000 side-to-side difference and manual Lachman test score at 1 year after surgery compared to that at baseline (all $p < 0.001$) (Table 3). There was a significant positive correlation between BMD at the distal femur and the single-leg hop distance of the injured limb at 1 year post-reconstruction ($r = 0.299, p = 0.031$).

4. Discussion

Our results showed that there was a statistically significant bone loss at the knee region and the hip. The BMD loss at the knee region was reversible at 1 year. The loss in BMD after ACL reconstruction was consistent with previous findings of BMD decrease at the knee and hip after ACL reconstruction [4,6]. Leppala et al. [4], reported 21%, 14%, 3% and 5% of interlimb BMD decrease at the distal femur, proximal tibia, trochanteric region and femoral neck, respectively, at 12 months post-operation. However, the bone loss did not return to normal in their study [4]. The BMD at the proximal tibia was also reported to be partially reversible at 24 months in another study [6].

Unlike the change of BMD at the knee region, the BMD loss at the hip was not reversible at 1 year post-operation, consistent with the findings of other studies investigating bone loss at the distal sites of injured limb [3–5]. Zerahn et al. [6], however, reported no significant difference in BMD at the calcaneus in 24 months of follow-up. As the follow-up time in this study was short (i.e. 1 year), longer follow-up is needed to determine if BMD at the hip would return to normal at a slower rate than that at the knee region. The rehabilitation program in our hospital focused on knee loading, but not hip loading. As many patients have instability post-ACL reconstruction related to poor proprioception and gluteal muscle control, future studies should address hip loading.

The transient loss of 5–10% of BMD around the knee has several implications for tendon graft to bone tunnel healing. First, the tunnel surface might become less stable for tendon–bone integration due to increased bone resorption. BMD decrease at the knee region might complicate revision surgery and might lead to degenerative joint diseases due to the production of inflammatory cytokines associated with bone resorption or even fracture due to the reduction in bone strength. Indeed, postoperative fracture is a serious, albeit, rare complication after ACL reconstruction. There were case reports of femoral condyle fractures [9,17,18] and proximal tibial fractures [19–21] after ACL reconstruction. All these injuries occurred through low energy trauma and most were sustained within the first several months after reconstruction. The bone tunnel and the graft harvest site might act synergistically as stress risers to reduce the bone strength of the proximal tibial and distal femur [8,9]. Particular care in the design of rehabilitation program or physical activity therefore must be taken in the first several months to protect the knee after surgery as a result of the inherent weakness of the bone around the knee. The effect of 2–5% decrease in BMD around the hip was likely to be mild as the patients in our study were young and active. Hence they were unlikely to suffer from fragility fracture as a result of the BMD decrease at the hip.

There was no significant difference in BMD loss at different sites with different surgical techniques. This indicated that different extent of trauma induced, different healing responses, locations and angles of tunnels in different surgical techniques had similar effects on BMD changes at the knee region and the hip.

Both early clinical and functional outcomes improved significantly at 1 year after operation. This was consistent with previous studies employing different surgical approaches [4–6]. However, there was

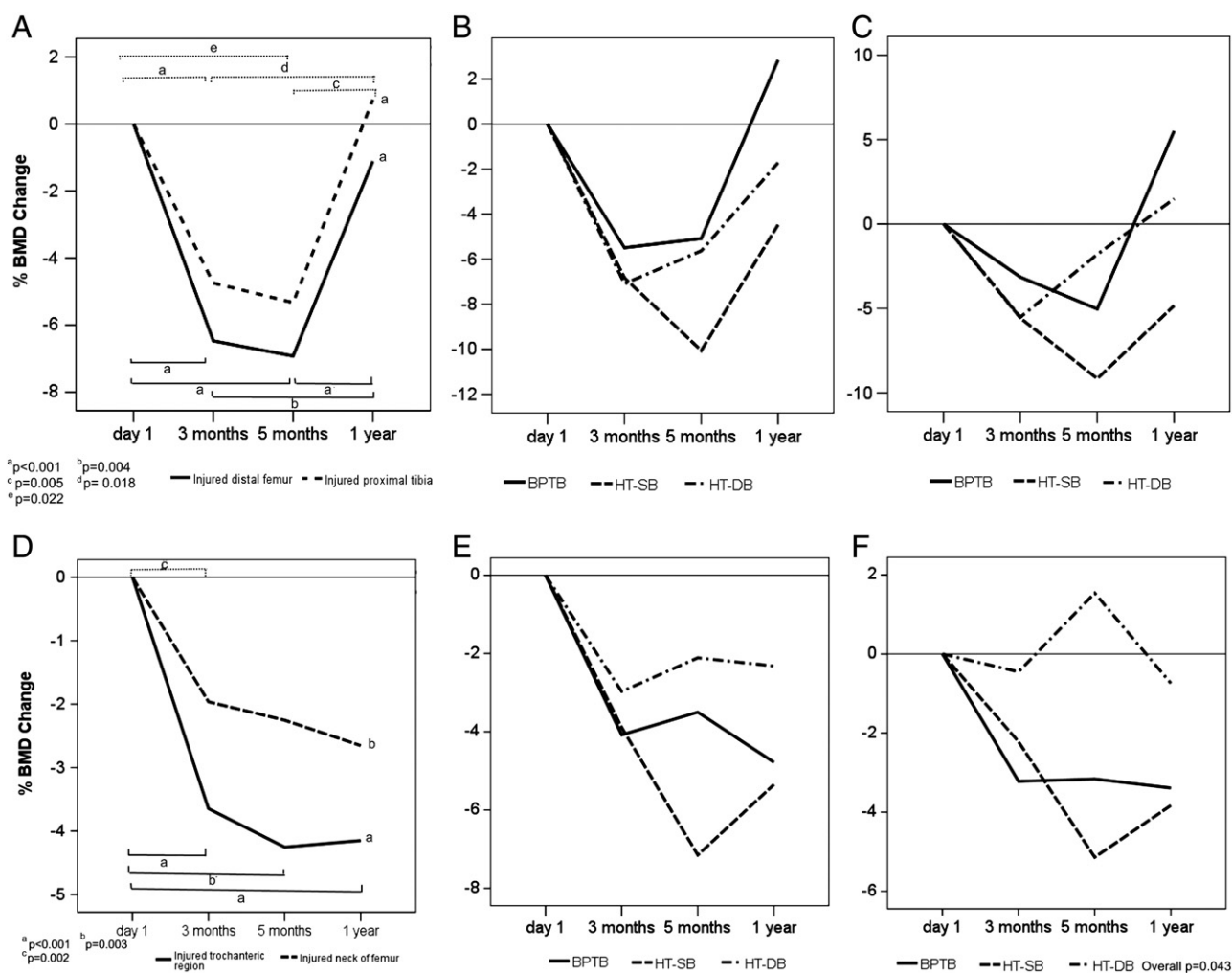


Fig. 3. Graphs showing the time-dependent percentage change in BMD in patients assigned to ACL reconstruction with BPTB, HT-SB or HT-DB. Panel (A) shows the pooled percentage change in BMD at distal femur (solid line) and proximal tibia (dotted line) while panels (B) and (C) show the percentage change in BMD at distal femur and proximal tibia, respectively, according to different surgical groups. Panel (D) shows the pooled percentage change in BMD at trochanteric region and femoral neck while panels (E) and (F) show the percentage change in BMD at trochanteric region and femoral neck, respectively, according to different surgical groups. In panels (A) and (D), the significant p value for all groups was indicated next to the line. The significant p values in post-hoc analyses were indicated at the bottom of each graph.

no significant difference in early clinical and functional outcomes among the three surgical techniques at 1 year. We reported significant positive correlation between BMD at the distal femur and the single-leg hop distance at 1 year, suggesting that BMD at the distal femur might positively affect the stability of the knee joint after reconstruction. Several authors reported that BMD correlated significantly with the functional scores of the injured knee after reconstruction [3,6]. Significant bone loss and decreased mechanical properties in the first 21 days after flexor tendon insertion site injury and repair were also reported in animal study, supporting the

relationship between bone loss and strength [1,2]. In this study, positive correlation was observed only in the distal femur but not in the proximal tibia. More research needs to be done to understand the mechanism. Longer-term study is also required to understand the association of BMD at the knee with the long-term functional and clinical outcomes of patients after ACL reconstruction.

There were several limitations in this study. Although block randomization was used to allocate the patients to different groups, we were unable to achieve equal sample size due to violation of protocol by the patients and loss to follow up with four patients not attending even one assessment, including baseline assessment, because of work. As all the patients were recruited from one hospital, the generalizability of the trial findings might be limited. We only included male patients in this study because of limited female patients in our hospital. Osteoporosis is a greater concern in women. The changes of BMD in female patients after ACL reconstruction and its potential impact on early and late functional and clinical outcomes of female patients need further research. Because of the limited sensitivity of DXA and we failed to put the knee and hip into the high-resolution pQCT machine for BMD measurement because of the small size of the measurement chamber, we were unable to achieve more precise local BMD measurement close to the tunnel region. Future studies should try to use other high-resolution bone imaging systems such as the XtremeCT for in vivo BMD measurement to elucidate the

Table 3

Table showing the changes of early clinical and functional outcomes at 1 year.

	n	Baseline	1 year	p value
IKDC	51	74.2 (13.3)	86.2 (10.1)	p<0.001
Lysholm score	51	84 (43–100)	95 (77–100)	p<0.001
One-leg hop test	50	0.6 (0.3)	0.8 (0.2)	p<0.001
Injured/non-injured				
KT-1000 (30 lbs) (mm)	51	3.7 (1.6)	2.1 (1.8)	p<0.001
Side-to-side difference				
Lachman test	51	3 (2–4)	1.5 (1–3)	p<0.001

Values for Lysholm score and Lachman test were presented as median (range) while values for the other parameters were represented as mean (SD).

changes of bone mass, density and bone micro-architecture close to the bone tunnel after ACL reconstruction.

5. Conclusion

The three surgical techniques were similar in transient bone loss at the knee region, irreversible bone loss at the hip, early clinical and functional outcomes up to 1 year post-reconstruction. BMD at the distal femur was positively associated with the single-leg hop distance at 1 year.

Conflict of interest statement

We declare that there is no conflict of interest in the publication of the research findings.

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